



GLOBAL ECONOMICS
GRADE: 10TH
MIDTERM TASK
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3 TERM
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Learning objective:

Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.

Assessment criteria

- C1. Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.
- C2. Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.
- C3. Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.
- C4. Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure may not receive credit. A minimum of 8 points is required to pass each criterion.

1. Weekend Market Stall

[C1-3 C2-3]

A craft cooperative must decide where to operate its stall for the weekend before knowing how strong customer demand will be. The cooperative can choose either a central stall or a neighbourhood stall, while weekend demand may turn out to be strong or weak.

If the cooperative chooses the central stall, it expects to earn a profit of fourteen thousand pounds when demand is strong and five thousand pounds when demand is weak. If it chooses the neighbourhood stall, it expects to earn a profit of ten thousand pounds when demand is strong and eight thousand pounds when demand is weak.

- a) (C1) Characterize the key elements of the decision-making situation. **[3]**
- b) (C2) Construct the payoff table. **[3]**

2. Urban Delivery Service

[C1-5 C3-10]

An urban logistics company provides transportation and delivery services across Bogotá. Before beginning daily operations, the company must decide how to organize the vehicles used to move packages between different parts of the city. It can either hire and operate its own dedicated van fleet or contract a third-party logistics provider to carry out the deliveries. Traffic conditions throughout the city may turn out to be relatively clear or heavily congested.

The payoff from the service depends on the number of deliveries successfully completed during the day, together with contributions and costs associated with both the chosen transportation arrangement and the traffic conditions. All monetary values are measured in thousand USD.

Hiring and operating its own van fleet provides a contribution of 0.7 thousand USD for each completed delivery and involves a fixed cost of 8 thousand USD for the day's operation. Contracting a third-party delivery service instead provides a contribution of 0.5 thousand USD for each completed delivery and involves a fixed cost of 13 thousand USD.

Traffic conditions also affect the financial result of providing the transportation service. When traffic across the city is relatively clear, each completed delivery contributes an additional 0.2 thousand USD, while the operation incurs an additional fixed cost of 4 thousand USD. When the city experiences heavy traffic, each completed delivery contributes an additional 0.1 thousand USD, while the op-

eration incurs a larger additional fixed cost of 6 thousand USD.

The number of deliveries that can be completed during the day also depends on both the transportation arrangement and the traffic conditions. With its own van fleet, the company expects to complete 80 deliveries when traffic is relatively clear and 55 when traffic is heavy. When using a third-party delivery service, it expects to complete 70 deliveries when traffic is relatively clear and 60 when traffic is heavy.

- a) (C1) Characterize the key elements of the decision-making situation. [5]
- b) (C2) Construct the payoff table. [10]

3. School Lunch Service [C1-3 C3-3]

A school enterprise must decide which lunch menu to offer before it knows how strong student demand will be. The enterprise can prepare either a hot-lunch menu or a sandwich menu, while demand may turn out to be high or low.

If the enterprise chooses the hot-lunch menu and demand is high, it expects total revenue of 24 thousand USD and total costs of 10 thousand USD. If demand is low, the same menu is expected to generate total revenue of 15 thousand USD and total costs of 9 thousand USD.

If the enterprise chooses the sandwich menu and demand is high, it expects total revenue of 21 thousand USD and total costs of 12 thousand USD. If demand is low, the sandwich menu is expected to generate total revenue of 18 thousand USD and total costs of 11 thousand USD.

- a) (C1) Characterize the key elements of the decision-making situation. [3]
- b) (C3) Calculate the profit, then construct the payoff table. [3]

4. Cold-Chain Distribution [C1-5 C3-10]

A food distributor must decide which refrigerated fleet to operate before it knows what demand and fuel conditions will be. The company can choose either a compact refrigerated fleet or a large refrigerated fleet. After this decision is made, market and fuel conditions may turn out to be favourable or adverse.

The number of delivery batches that can be handled depends on both the fleet selected and the conditions that occur. With the compact refrigerated fleet, the distributor expects to handle 800 delivery batches under favourable conditions and 300 under adverse conditions. With the large refrigerated fleet, it expects to handle 1,100 delivery batches under favourable conditions and 400 under adverse conditions.

Revenue from each 100 delivery batches depends partly on the fleet selected. The compact refrigerated fleet generates \$12,000 per 100 batches, while the large refrigerated fleet generates \$16,000 per 100 batches. Revenue is also affected by the conditions that occur. Favourable conditions add \$3,000 of revenue per 100 batches, while adverse conditions add \$1,000.

The fleet choice also creates variable costs for each 100 delivery batches. The compact refrigerated fleet has a variable cost of \$5,000 per 100 batches, while the large refrigerated fleet has a variable cost of \$7,000 per 100 batches. Conditions create an additional variable cost as well. Favourable conditions add \$1,000 of cost per 100 batches, while adverse conditions add \$4,000.

In addition to these variable amounts, the fleet selected generates fixed revenues and fixed costs. Choosing the compact refrigerated fleet provides \$16,000 in fixed revenue and creates \$20,000 in fixed costs. Choosing the large refrigerated fleet provides \$25,000 in fixed revenue and creates \$35,000 in fixed costs.

The conditions that occur also generate fixed financial effects. Favourable conditions provide \$12,000 in additional fixed revenue and create \$6,000 in additional fixed costs. Adverse conditions provide \$4,000 in additional fixed revenue and create \$14,000 in additional fixed costs.

- a) (C1) Characterize the key elements of the decision-making situation. [5]
- b) (C3) Calculate the profit, then construct the payoff table. [10]

5. Mobile Produce Shop [C4-2]

A produce seller must locate a mobile shop at the business district (A_1) or the residential district (A_2). Demand may be high (S_1) or low (S_2). Profits, in thousand USD, are:

Alternative	S_1	S_2
Business district	48	12
Residential district	35	24

- a) Apply Maximax and state the selected location. **[1]**
b) Apply Maximin and state the selected location. **[1]**

6. Warehouse Technology **[C4-8]**

A wholesaler must choose one of four warehouse systems before market and supply conditions are known. The possible states are strong demand (S_1), steady demand (S_2), and

weak demand with delays (S_3). The profits, in thousand USD, are:

Alternative	S_1	S_2	S_3
Automated system	96	52	18
Modular system	78	61	42
Robotic system	88	48	30
Flexible system	72	64	50

- a) Apply Maximax and state the selected system. **[4]**
b) Apply Maximin and state the selected system. **[4]**